

# 江南大学 2022-2023-1 学年度《概率论与数理统计 II》期末考试卷 参考答案和评分标准

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| <p>一、填空题<br/>(每空 5 分, 共 50 分)</p> <p>1. <math>ABC\bar{C}</math></p> <p>2. 0.6</p> <p>3. 0.25</p> <p>4. <math>2/\bar{X}</math></p> <p>5. [4.10, 42.60]</p> <p>6. <math>\bar{X} - 1</math></p> <p>7. <math>E(1/3)</math></p> <p>2</p> <p>8. <math>4/5</math></p> <p><math>3/5</math></p> | <p>二、计算题</p> <p>1.</p> <p>定义随机事件 <math>N</math>、<math>U</math> 分别为抽到的是正常币、错铸币,</p> <p><math>A_i</math> 为第 <math>i</math> 次抛掷正面朝上. ....(4 分)</p> <p>则 <math>P(N) = 0.5, P(U) = 0.5</math>,</p> <p><math>P(A_1A_2 N) = 0.5 * 0.5 = 0.25, P(A_1A_2 U) = 1 * 1 = 1</math></p> <p>这枚硬币是正常币的概率为</p> $P(N A_1A_2) = \frac{P(A_1A_2 N)P(N)}{P(A_1A_2 N)P(N) + P(A_1A_2 U)P(U)}$ $= 0.2$ <p>.....(5 分)</p>                                                                                                                       | <p>3.</p> <p>1. 由 <math>\int_{-\infty}^{+\infty} kx^2 dx = \int_0^1 kx^2 dx = 1</math> 得 <math>k = 3</math> .....(4 分)</p> <p>2. <math>F(x) = \int_{-\infty}^x f(t)dt = \begin{cases} 0 &amp; x \leq 0 \\ x^3 &amp; 0 &lt; x \leq 1 \\ 1 &amp; x &gt; 1 \end{cases}</math> .....(4 分)</p> <p>3. 令 <math>Y = X^2</math>,</p> <p>当 <math>y \leq 0, F_Y(y) = 0</math>; 当 <math>y &gt; 1, F_Y(y) = 1</math>; 当 <math>0 &lt; y \leq 1</math>,</p> $F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P\left(X \leq y^{\frac{1}{2}}\right) = F_X\left(y^{\frac{1}{2}}\right) = y^{\frac{3}{2}}$ $P(X^2 \leq 1) = P(Y \leq 1) = 1$ <p>.....(5 分)</p>                                                                             |
|                                                                                                                                                                                                                                                                                        | <p>2.</p> <p>(1) 提出假设 <math>H_0: \mu = \mu_0 = 10.5; H_1: \mu \neq \mu_0</math></p> <p>选择检验统计量 <math>T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t(n-1)</math> .....(3 分)</p> <p>(2) 对于显著性水平 <math>\alpha = 0.05, t_{0.025}(17) = 2.11</math>,</p> <p><math>W = (-\infty, -2.11] \cup [2.11, +\infty)</math> .....(3 分)</p> <p>(3) 计算统计量, <math>T = \frac{10.5 - 11.1}{\frac{2}{\sqrt{18}}} \cong -1.27</math> .....(3 分)</p> <p>(4) 做出判断: 因为 <math>T \notin W</math>, 所以接受 <math>H_0</math>, 认为 “未发生变化” .....(3 分)</p> | <p>4.</p> <p>(1) <math>F_Z(z) = P(Z \leq z)</math></p> <p><math>Z \leq 0</math> 时, <math>F_Z(z) = 0</math>; <math>0 &lt; Z \leq 1</math> 时, <math>F_Z(z) = \frac{\pi z^2}{\pi 1^2} = z^2</math>; <math>Z &gt; 1</math> 时, <math>F_Z(z) = 1</math>;</p> $f_Z(z) = F'_Z(z) = \begin{cases} 2z, & 0 < z \leq 1 \\ 0, & \text{others} \end{cases}$ <p>.....(8 分)</p> <p>(2) <math>F_T(t) = P(T \leq t) = P\left(\frac{Y}{X} \leq t\right) = \begin{cases} P(Y \geq Xt), &amp; X &lt; 0 \\ P(Y \leq Xt), &amp; X &gt; 0 \end{cases}</math></p> $= 2 \left[ \frac{1}{2} \left( \arctan t + \frac{\pi}{2} \right)^2 \right] / \pi 1^2 = \frac{\arctan t}{\pi} + \frac{1}{2}, -\infty < t < +\infty$ <p>.....(5 分)</p> |

